

Failure behaviour of carbon fibre support layer in foldable OLED modules

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Abstract

The support layer of foldable OLED module is a key component that determines the thickness, weight and bending performance of the screen, and lightweight and high-strength carbon fibre as the support layer material has become the mainstream trend. The design of periodically patterned apertures is the main means to improve the bendability of the substrate, but there is a risk of fracture failure in the apertures under actual working conditions. In order to investigate the failure mechanism of the carbon fibre support layer. In this paper, [90/0/90] fibre ply specimens are prepared for quasi-static tensile tests and a progressive damage model of the laminate is established using the Vumat subroutine. The failure mechanism of the backing layer is analysed by numerical simulation and experiment, and the fracture morphological characteristics and microscopic failure mode of the open hole region are observed by scanning electron microscope (SEM). The results show that the damage extends from the region of maximum stress concentration around the hole to the free edge, the initial damage is borne by the matrix in the 90° lay-up direction, and the final failure behaviour is dominated by the fibre layer in the 0° lay-up direction. The simulation results are in agreement with the experimental results and confirm the correctness of the progressive damage model.

Author Keywords

Foldable OLED Module; Carbon Fibre Support Layer; Damage Evolution; Failure Analysis.

1. Introduction

The ideal support plate material should have the advantages of high specific strength and stiffness, good bending performance, etc., which can well solve the problem of high reliable bending stability^[1-2], and lightweight high-strength carbon fiber as the support plate material has become the development trend of the next generation of OLED folding screen^[3-4]. In order to ensure the bending performance and support performance of the support plate

at the same time, it is necessary to pattern the bending area with perforation design, and laser perforation will inevitably cut off the local fiber, which will reduce the mechanical properties of the support plate to a certain extent^[5-6].

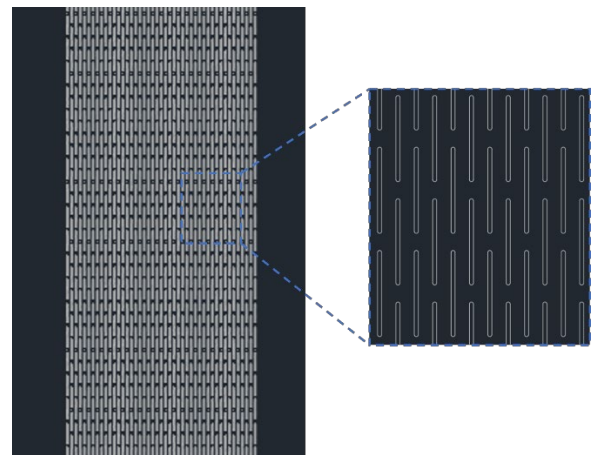


Fig.1. Flexible OLED foldable screen patterned support plate structure

Therefore, in actual working conditions, the support plate is prone to fiber breakage, matrix cracking and other damage failure. However, the existing studies on fatigue damage and life prediction of carbon fiber composites mainly focus on intact carbon fiber laminates or single-hole cases, and there is a lack of corresponding theoretical and experimental studies on patterned processed porous carbon fiber laminates due to their complex structure and stress distribution. Therefore, in this paper, ABAQUS combined with subroutine development is used to simulate the fracture process of carbon fiber laminates, so as to understand the stress distribution and fracture mechanism in the perforated region of the laminates.

2. Modeling

2.1 Theoretical model

The damage forms of carbon fiber composites include fiber fracture, interlaminar delamination, and matrix fiber stripping, etc. Therefore, the three-dimensional Hashin failure criterion is used as

the damage initiation criterion. In ABAQUS, the two-dimensional Hashin failure criterion and various damage evolution criteria are integrated, but the three-dimensional intrinsic model of the composite material is missing, so it needs to be developed twice by using the VUMAT subroutine. In the VUMAT subroutine, the stress tensor needs to be updated based on the material's intrinsic relationship in conjunction with the current strain state. The solver calculates the nodal forces from the stress values and thus solves the mechanical response of the structure. Therefore, it is necessary to derive the intrinsic relationship of the material. Carbon fiber material is a transverse anisotropic material, and according to the classical laminate theory, the intrinsic relationship of the carbon fiber composite material when it is lossless is:

$$\begin{bmatrix} \sigma_1 \\ \sigma_2 \\ \sigma_3 \\ \sigma_4 \\ \sigma_5 \\ \sigma_6 \end{bmatrix} = \begin{bmatrix} Q_{11} & Q_{12} & Q_{13} & 0 & 0 & 0 \\ Q_{12} & Q_{22} & Q_{23} & 0 & 0 & 0 \\ Q_{13} & Q_{23} & Q_{33} & 0 & 0 & 0 \\ 0 & 0 & 0 & Q_{44} & 0 & 0 \\ 0 & 0 & 0 & 0 & Q_{55} & 0 \\ 0 & 0 & 0 & 0 & 0 & Q_{66} \end{bmatrix} \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \varepsilon_3 \\ \varepsilon_4 \\ \varepsilon_5 \\ \varepsilon_6 \end{bmatrix} = [Q^d] \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \varepsilon_3 \\ \varepsilon_4 \\ \varepsilon_5 \\ \varepsilon_6 \end{bmatrix}, \text{ which } \begin{cases} Q_{11} = E_1(1 - \nu_{23}\nu_{32})\Delta \\ Q_{22} = E_2(1 - \nu_{13}\nu_{31})\Delta \\ Q_{33} = E_3(1 - \nu_{12}\nu_{21})\Delta \\ Q_{12} = E_1(\nu_{21} + \nu_{31}\nu_{23})\Delta \\ Q_{13} = E_1(\nu_{31} + \nu_{21}\nu_{32})\Delta \\ Q_{23} = E_2(\nu_{32} + \nu_{12}\nu_{31})\Delta \\ Q_{44} = G_{23} \quad Q_{41} = G_{31} \quad Q_{42} = G_{12} \end{cases}$$

$$\Delta = \frac{1}{1 - \nu_{12}\nu_{21} - \nu_{23}\nu_{32} - \nu_{31}\nu_{13} - 2\nu_{21}\nu_{32}\nu_{13}}, \text{ the Poisson's ratio}$$

$$\text{relationship is defined as } \frac{E_i}{E_j} = \frac{\nu_{ij}}{\nu_{ji}}$$

The progressive damage model is a theoretical model used to analyze and predict the internal damage from the incipient, expansion to the final failure of the material in the process of subjecting to complex loads^[7]. The support plate is a gradual process from damage initiation to complete failure, so the bilinear model in the progressive damage model is used for the damage evolution, and the normal cohesion intrinsic model is taken as an example, and the stress-strain relationship is shown in Fig. 2 below:

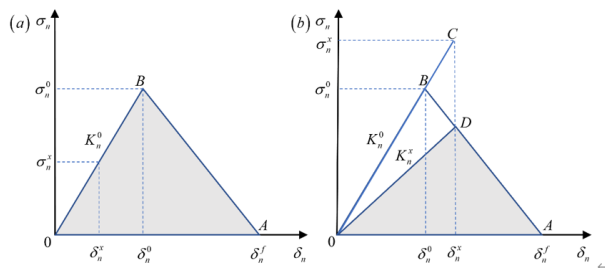


Fig. 2. Normal cohesion principal model stress-strain relationship

When the separation displacement is greater than δ_n^0 , the model enters the damage evolution stage and the model produces damage. As the separation displacement increases, the cohesion decreases until it drops to 0. A normalized monotonic damage variable D_n ,

is introduced into the cohesion model, with a value of D_n of 0 indicating that the model produces no damage, 1 indicating that the model is fully damaged, and values in between indicating that the cohesion model is in varying degrees of damage. The damage variable can be expressed as:

$$D_n = \frac{\delta_n^f (\delta_n^x - \delta_n^0)}{\delta_n^x (\delta_n^f - \delta_n^0)} \quad \delta_n^0 \leq \delta_n^x \leq \delta_n^f$$

With the introduction of the damage variable, the damage stiffness matrix becomes:

$$\begin{bmatrix} d_1 Q_{11} & d_2 Q_{12} & d_3 Q_{13} & 0 & 0 & 0 \\ & d_2 Q_{22} & d_3 Q_{23} & 0 & 0 & 0 \\ & & d_3 Q_{33} & 0 & 0 & 0 \\ & & & d_4 Q_{44} & 0 & 0 \\ & & & & d_5 Q_{55} & 0 \\ & & & & & d_6 Q_{66} \end{bmatrix}, \text{ which } \begin{cases} d_1 = 1 - df \\ d_2 = (1 - df)(1 - dm) \\ d_3 = (1 - df)(1 - smt * dmt)(1 - smc * dmc) \end{cases}$$

2.2 Subprogram Development Process

User material subroutine secondary development interface is provided in ABAQUS, in which UMAT subroutine is suitable for implicit solving and VUMAT is suitable for explicit solving. In this paper, ABAQUS/EXPLICIT solver is used to write the VUMAT subroutine with the following flowchart:

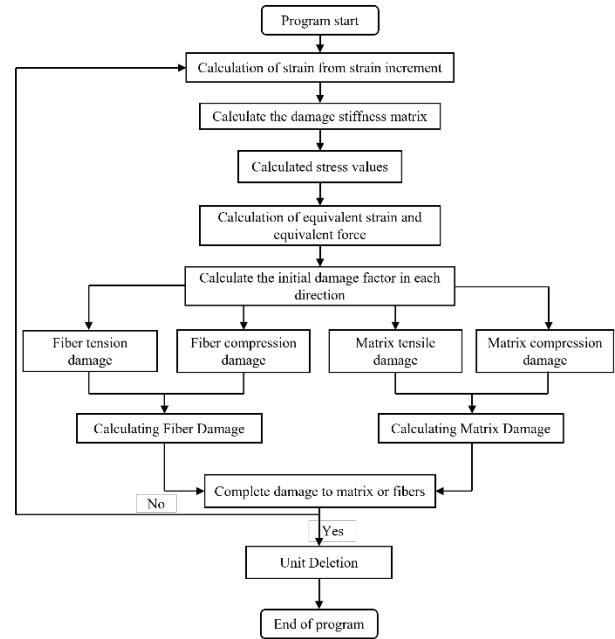


Fig. 3. Flowchart of subroutine solution

2.3 Experimental and numerical simulation

According to GB/T1447-2005 standard for plywood sampling, the sample size of 160mm long×20mm wide×0.175mm thick (thickness direction of 0.05mm, 0.115mm, 0.05mm), the fiber layout direction is [90/0/90]. The tensile experiment was carried out on an electronic dynamic and static fatigue testing machine, the lower end

of the specimen was fixed during the test, and the upper end was driven by the testing machine to stretch upward at a speed of 1mm/min until the plywood was pulled off, the whole process to keep the specimen and the chuck always in a parallel state, and the specimen was loaded as shown in the following Fig. 4. At the same time, the plywood model was created in ABAQUS to give the simulation analysis.



Fig. 4. Loading diagram of laminated plywood specimen

3. Results and Discussion

3.1 Experimental and simulation analysis

Fig. 5 shows the force-displacement curves of carbon fiber plywood under tensile load, and each curve has a good similarity in trend. At the initial stage, the stress-strain of the plywood under tensile load is linear, and the elastic deformation occurs in the pattern region of the plywood, which corresponds to the initial stiffness of the carbon fiber when it is undamaged; with the increasing load, it enters into the transition stage, and the 90° direction of the fiber layer undergoes the local plastic yielding and the delamination damage of the ply; when the load continues to increase, the slope of the curve decreases to arrive at the post-yield stage, and the 0° direction of the fiber layer bears the main load, and the tensile strength of the plywood gradually reaches the maximum. The fiber layer bears the main load, and the tensile strength of the plywood gradually reaches the maximum value; subsequently, the specimen issues a fracture crunching sound at the hole circumference, and the stress decreases rapidly, and the plywood then fails by fracture.

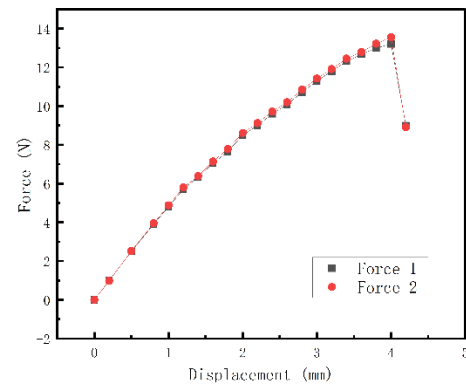


Fig. 5. Stress-strain curve of plywood

The numerical simulation results of tensile fracture failure of carbon fiber plywood are shown in Fig. 6 below. From the figure, it can be seen that the plastic deformation of the fiber ply under tensile loading occurs, and the cracks first appear at the edge of the pattern holes, because these regions are prone to stress concentration, which makes the actual stresses on the fiber and the matrix much higher than the average stresses, and thus it is more likely to reach the strength limit of the material, which triggers the tensile failure of the fiber and the matrix. Subsequently, the cracks gradually expand towards the region of maximum stress concentration, but the crack expansion is not completely perpendicular to the tensile direction, probably due to the inconsistent coordination of deformation between different fiber layers and the lack of constraints in the undeformed direction.

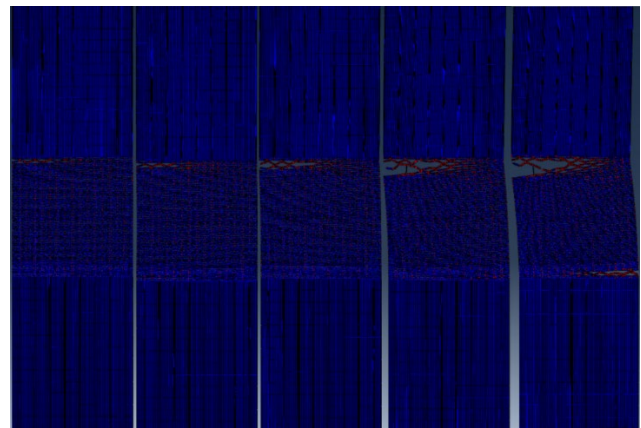


Fig. 6. Damage process and stress distribution of specimen under 1mm/min loading rate

In terms of delamination, the fiber direction of 90° plywood layer is perpendicular to the loading direction, the bearing capacity is weak, and it mainly relies on the matrix to withstand the tensile damage. 90° layer firstly shows damage around the holes and

expands inward, the damage degree is serious, as shown in Fig. 7 below.

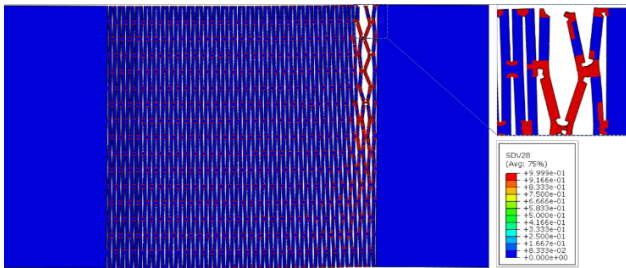


Fig.7. Matrix tensile damage results in 90° layup direction

Since the 0° ply is the main load-bearing ply, the 0° ply still has load-bearing capacity after damage occurs to the 90° ply. 0° ply damage first occurs at the free edge and around the hole, and then expands in the direction of the width of the supporting plate reinforcement, and ultimately the damage extends through the entire reinforcement, and the plywood loses its load-bearing capacity.

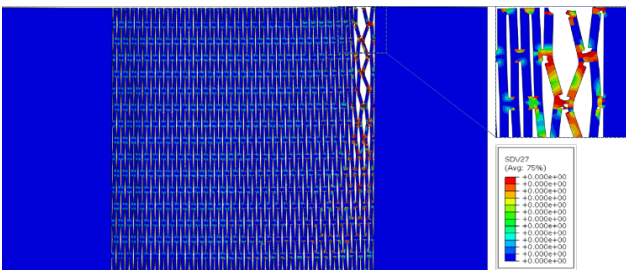


Fig.8. Fiber tensile damage results in 0° layup direction

The reason why damage is more likely to occur in the 90° paving direction is that the tensile and compressive strengths of the matrix resin are much smaller than those of the carbon fibers; at the same time, although the carbon fibers themselves have a high longitudinal strength, the strength of the fibers in the 90° paving direction, i.e., in the transverse direction, is relatively low because the carbon fiber filaments in the longitudinal direction are composed of chains of carbon atoms that are highly oriented and are able to withstand tensile stresses well, but in the transverse direction the bonding force between atoms is relatively weak. The bonding between atoms is relatively weak, and the ability to resist stretching is limited. 0° layup direction, the carbon fiber filament has a large tensile strength, only when the tensile stress of the support plate exceeds the tensile strength, the atomic bonds within the fiber filament break, which is manifested as the destruction of the crystalline structure of the fiber in the microcosm, and in the macrocosm as the failure of the fiber in tensile failure.

Micro-mechanism analysis

In order to further understand the microstructural characteristics such as fiber distribution, orientation, matrix state and interfacial bonding within the carbon fiber laminates, and to analyze the micro-morphology of the fractured specimens, the fractured specimens were sampled by scanning electron microscope (SEM) using sirion200, and their micro-morphology is shown in the following Fig.9.

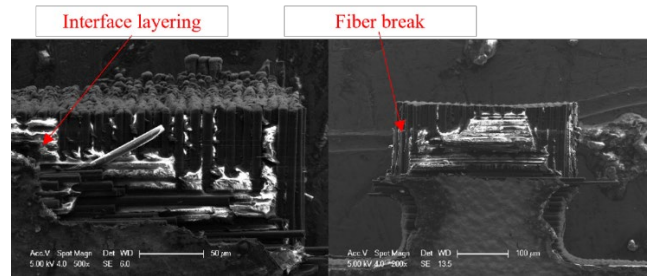


Fig. 9. Microcosmic section of specimen

From the microcosmic cross-section can be seen, under the action of tensile load, 0° transverse fiber layer has been completely fractured, most of the epoxy resin in the tensile fracture moment of the impact generated by the fragmentation and fall off, only a small amount of debris is still bonded to the fiber surface, some of the fibers are also pulled out; In addition to the fracture, there are a small portion of the fibers in the internal fracture, probably due to the reasons of the plywood lay-up method, so that some of the fibers after the bonding of resin density. It may be due to the reason of plywood layup, which makes the density of some fibers after resin bonding uneven, and produces tension difference, which leads to the fracture of internal fibers first. At the same time, it can also be seen that there is fiber delamination damage between the layers, on the one hand, due to the significant anisotropy of the carbon fiber filaments themselves, in the direction of the fibers with high strength and stiffness, but in the direction perpendicular to the fibers, especially in the direction of the interlaminar direction, the mechanical properties of the relatively weak. When subjected to tensile load, the shear resistance of the interlayers is insufficient, leading to relative sliding and separation of the layers, which in turn triggers delamination; on the other hand, the unreasonable lay-up angle and sequence lead to uncoordinated deformation between the layers of the plywood when subjected to force. The difference in lay-up angle between neighboring layers is too large, and the deformation trend of each layer is different in the tensile process, resulting in the mismatch of the stiffness of each layer and thus

generating a large interlayer shear force, and delamination occurs when the shear force exceeds the bonding strength of the interlayer.

4. Conclusion

This paper reveals the stress distribution and damage mechanism in the pattern region of carbon fiber plywood during the fracture process. There is an obvious stress concentration phenomenon in the patterning region under tensile load, which is manifested as the maximum stress at the hole edge perpendicular to the tensile direction and the minimum stress at the hole edge along the tensile direction. During the tensile fracture process, the damage and microcracks are first produced at the free edge and hole perimeter, and then the cracks keep expanding along the width of the tendon, and the fracture location is located at the junction of the patterning hole and the plate, and the simulation results coincide with the experimental results. In terms of delamination, the matrix tensile failure occurs first in 90° layer, and then fiber tensile failure occurs in 0° layer, and 0° layer is the main load-bearing layout; at the same time, there are fiber pullout and interface delamination phenomenon in the fracture process.

5. Acknowledgements

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6. References

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